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Thoracic: Lung

A randomized study of cryoablation of intercostal nerves in patients undergoing minimally invasive thoracic surgery

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Commentator Discussion: A randomized study of cryoablation of intercostal nerves in patients undergoing minimally invasive thoracic surgery
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Abstract

Objectives

Minimally invasive thoracic surgery can cause significant pain, and optimizing pain control after surgery is highly desirable. We examined pain control after intercostal nerve block with or without cryoablation of the intercostal nerves.

Methods

This was a randomized study (NCT05348447) of adults scheduled for a minimally invasive thoracic procedure. Each intercostal space near the incision site was injected with lidocaine and bupivacaine with epinephrine (standard of care). The cryoanalgesia group also had 5 to 6 intercostal nerves ablated. The primary outcome was the amount of narcotics (in morphine milligram equivalents taken during the postoperative hospital stay and the first 2 weeks postdischarge. Secondary outcomes were incentive spirometry volume and pain scores in the hospital and pain and neuropathy scores at 2 weeks.

Results

Our final cohort contained 103 patients (52 standard of care and 51 cryoanalgesia). There were no differences between the treatment groups in morphine milligram equivalents administered during the hospital stay (44.9 vs 38.4 mg), total morphine milligram equivalents at 2 weeks (108.8 vs 95.2 mg), or pain assessed by visual analog scale on postoperative day 1 (3.8 and 3.3), postoperative day 2 (2 and 3.5), or 2 weeks (2 and 3.5) for standard of care and cryoanalgesia group patients, respectively. The decrease in incentive spirometry during the postoperative period was not significantly different between the 2 groups. Patients in the cryoanalgesia group had higher neuropathy scores (8 vs 13; P=.019) 2 weeks after surgery.

Conclusions

In this randomized study, cryoanalgesia did not decrease postoperative pain or narcotic requirements. Cryoanalgesia increased neuropathic pain 2 weeks after surgery.

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Methods

This was a prospective, randomized study registered in ClinicalTrials.gov ↗ as NCT05349447. The trial opened in April 2022 and closed in January 2024. The Institutional Review Board of the Allegheny Health Network approved the study (No. 2021-238; approved November 9, 2021), and all patients gave informed consent for the research. Atricure Inc provided a research grant for this

study. The investigators designed the trial and analyzed the results freely, without interference from the funding …

Results

We recruited 110 patients for the study. Before randomization, 3 patients decided to withdraw from the study. After randomization but before treatment, four patients (2 randomized to standard of care and 2 randomized to cryoanalgesia) were excluded because 2 decided not to pursue surgery, and 2 withdrew consent for the study. Our final cohort contained 103 patients (Figure1). The study groups were similar in age, sex, body mass index, and preoperative lung function (Table 1). Most patients …

Discussion

In a randomized study comparing the combination of intercostal nerve block and cryoanalgesia with intercostal nerve block alone, we did not find significant differences in narcotic use or pain after minimally invasive thoracic surgery. As measured by the LANSS questionnaire, patients undergoing cryoanalgesia had significantly higher neuropathic pain scores compared with controls. Although the incentive spirometry on days 1 and 2 was higher in the experimental group, the difference between pre- …

Conclusions

In this randomized study of patients undergoing minimally invasive thoracic surgery, the use of cryoanalgesia with bupivacaine and lidocaine did not further decrease pain or reduce narcotic use during the early postoperative period. There was no change in respiratory function, but more neuropathy was reported at the first postoperative visit. We are currently investigating if ablating fewer intercostal nerves for less time may provide a decrease in narcotic use after minimally invasive thoracic …

Conflict of Interest Statement

Dr Weksler is a proctor for Intuitive Surgery and a speaker for AstraZeneca and Merck. All other authors reported no conflicts of interest.

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IRB-approval: The IRB of the Allegheny Health Network approved the study (IRB number 2021-238, approved on November 9, 2021).

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